

Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #08: Digital Campus Library Reservation Gateway

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1. **Architecture used** : best architecture is layered , else we can use microservices and client-server architecture too but they wouldn't be that efficient as layered architecture.

Why layered ? The system has several clearly separable responsibilities:

Presentation → Business Logic → Data Access → Database/External Services

2. Functional Requirements

ID	Type	Requirement Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow users to search the library catalog using an ISBN.	High	Pass: Matching book details are displayed. Fail: Incorrect or missing results are displayed.	Enables users to quickly locate physical books.
FR-002	Functional	The system shall allow users to place a hold on an unavailable book.	High	Pass: A valid hold is added to the queue. Fail: Hold is rejected when the user reaches the limit.	Allows users to reserve currently borrowed books.
FR-003	Functional	The system shall maintain hold requests in FIFO queue order.	High	Pass: Requests are served in submission order. Fail: A later request is served first.	Ensures fair allocation of returned books.
FR-004	Functional	The system shall notify the next user when a held book becomes available.	Medium	Pass: Correct user receives an availability notification. Fail: Notification is missing or sent incorrectly.	Keeps users informed without manual staff intervention.
FR-005	Functional	The system shall record and track overdue-fine	High	Pass: Fine details are correctly recorded. Fail: Incorrect or	Provides accurate tracking

ID	Type	Requirement Description	Priority	Acceptance Criteria	Rationale
		transactions for borrowed books.		missing fine information is stored.	of overdue transactions.

3. Non-Functional Requirements

ID	Type	Requirement Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	The system shall return ISBN search results within 2 seconds under normal load.	High	Pass: 95% of searches finish within 2 seconds. Fail: More than 5% exceed 2 seconds.	Provides fast and responsive catalog searches.
NFR-002	Security	The system shall restrict access to member, reservation, and fine data based on user roles.	High	Pass: Users access only authorized data and functions. Fail: Unauthorized data or functions are accessible.	Protects sensitive student and transaction information.

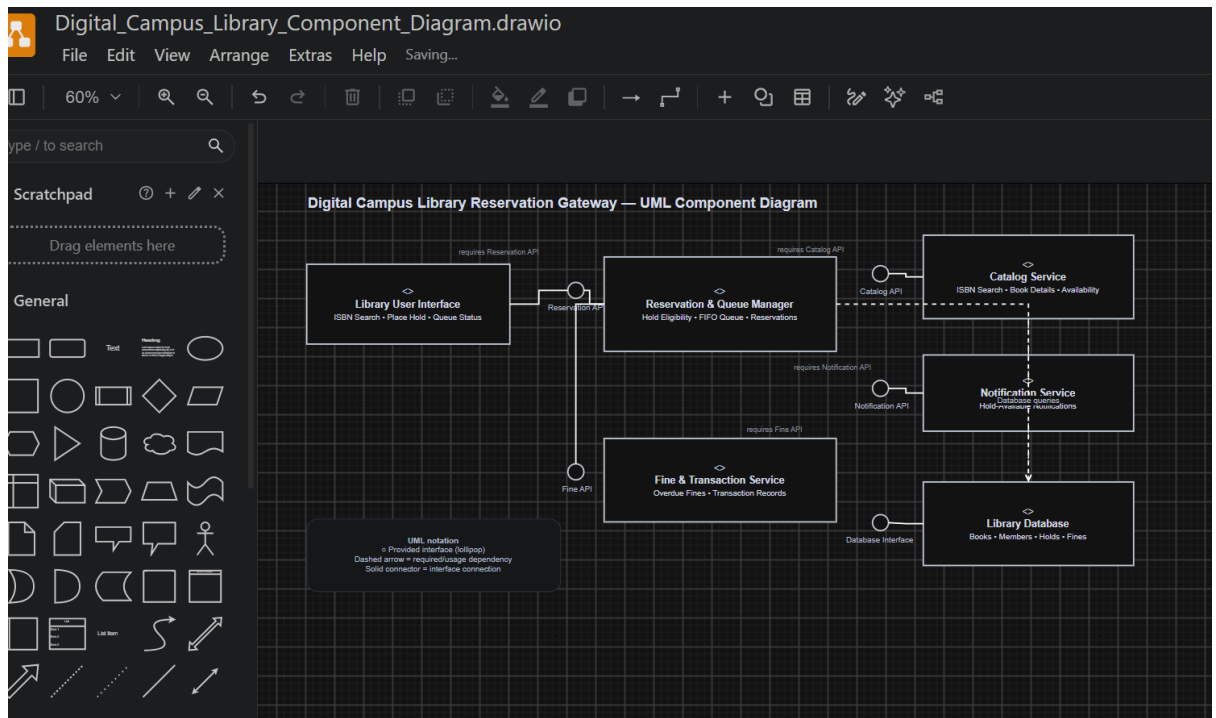
4. UML Use-Case Diagram

Two actors are:

- **Student Member**
- **Head Librarian**

The main use cases can be:

- Search ISBN Catalog
- View Book Availability
- Place Hold Request
- View Queue Position
- Receive Hold Notification
- Manage Holding Queue
- Manage Book Catalog
- Track Overdue Fine Transactions
- View Fine Transactions



5. Core Use Case Flow Specification :

Use Case: Place Hold Request

Use Case ID: UC-03

Use Case Name: Place Hold Request

Primary Actor: Student Member

Preconditions :

1. Student Member is registered in the library system.
2. Student Member has successfully authenticated.
3. The requested ISBN exists in the library catalog.
4. The requested physical book is currently unavailable/borrowed.
5. The Student Member has not reached the maximum permitted number of holds.

Postconditions

Success:

1. A hold request is created.
2. Student Member is added to the appropriate FIFO queue.
3. The student's queue position is displayed.
4. The reservation is stored in the system.

Failure:

1. No hold request is created.
2. The student's existing reservations remain unchanged.
3. The user receives an appropriate error message.

Main Success Scenario

1. Student Member logs into the Digital Campus Library Reservation Gateway.
2. Student searches for a book using its ISBN.
3. System searches the physical ISBN catalog.
4. System displays the matching book and its availability.
5. Student selects the unavailable book and chooses **Place Hold**.
6. System checks the student's hold eligibility.
7. System verifies that the student has not exceeded the maximum allowed holds.
8. System creates a hold request.
9. System places the student at the end of the FIFO holding queue.
10. System calculates the student's queue position.
11. System displays confirmation and queue position to the Student Member.
12. System stores the reservation information.

6.Component Diagram

Component 1 — Library User Interface

Handles:

- ISBN search
- Book availability display
- Hold requests
- Queue position display
- Notifications

Component 2 — Reservation & Queue Manager

Handles:

- Hold creation
- FIFO queue management
- Hold eligibility
- Queue position
- Reservation status

Component 3 — Catalog Management Service

Handles:

- ISBN search
- Book information
- Physical copy availability
- Catalog queries

Component 4 — Notification Service

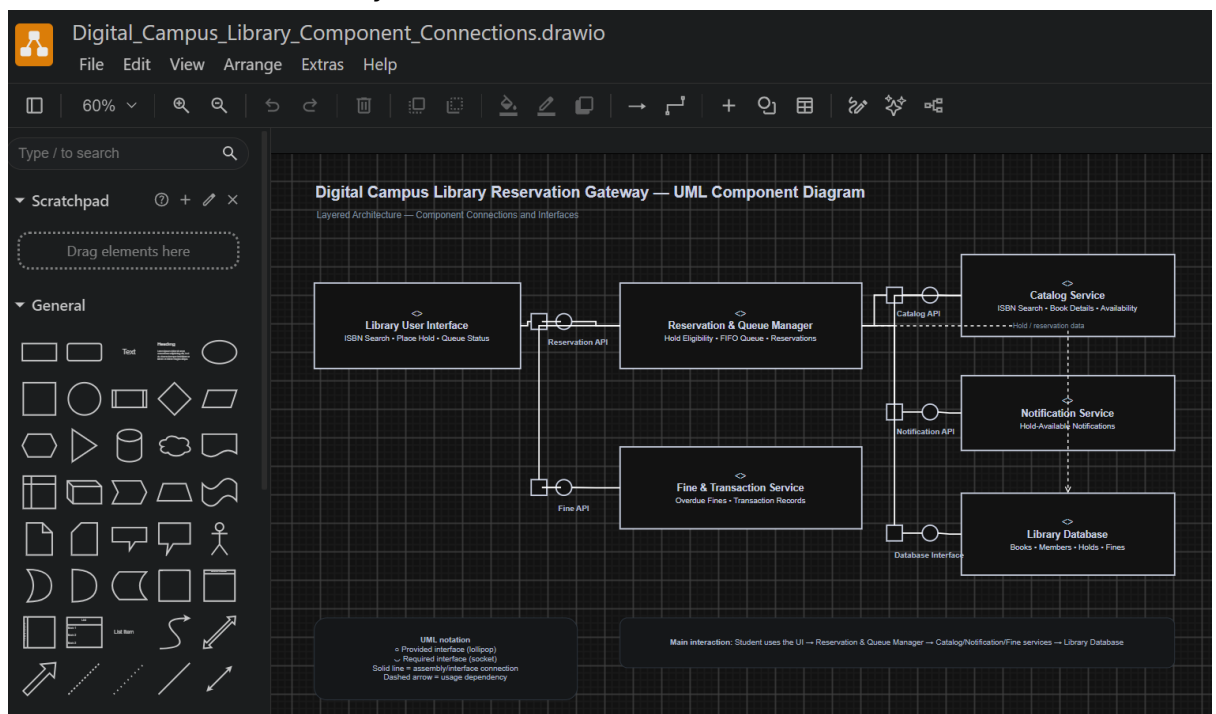
Handles:

- Hold-available notifications
- Reservation notifications
- Notification delivery status

Component 5 — Fine & Transaction Service

Handles:

- Overdue fine records
- Fine transactions
- Member transaction history



8. One-Page Architecture:

Architecture Selection: Layered Architecture

We chose Layered Architecture for the Digital Campus Library Reservation Gateway.

The first reason for selecting Layered Architecture is that the system contains several clearly separated responsibilities, including ISBN catalog searching, reservation and FIFO queue management, automated notifications, and overdue-fine transaction tracking. Separating the presentation, business-logic, and data-access responsibilities makes the system easier to understand, maintain, test, and modify without affecting unrelated functionality.

The second reason is that the system is a centralized university library application rather than a collection of highly independent business services. A layered structure allows reservation and queue rules to remain in the business layer while catalog and fine

information can be managed through a separate data layer. This is appropriate for the relatively structured operations of a campus library system.

Security Advantage:

The architecture allows authentication and authorization to be handled at the presentation/service boundary while sensitive member and fine information remains behind the business and data-access layers. Role-based access can ensure that Student Members cannot access administrative library functions reserved for the Head Librarian.

Performance Benefit:

The catalog and reservation data can be accessed through dedicated data-access components, while frequently requested ISBN and availability information can be efficiently queried or cached. This reduces unnecessary processing and helps the system respond quickly to student searches and reservation requests.

Conclusion:

Layered Architecture provides a simple and maintainable structure for the Digital Campus Library Reservation Gateway while supporting the required reservation, queue, notification, catalog, and fine-management functionality.

9. Checkout github link : [Himani-PES2UG24CS194 \(Himani-194\)](#)